# Compute Core — Mathematical Description

5-Class Cardiac Arrhythmia Classifier, RBF-SVM ASIC


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The `svm_compute_core` implements two sequential subfunctions per support vector: a **distance function** (`dist`) that computes the squared Euclidean distance between an input feature vector and a support vector, and a **kernel function** (`horner`) that evaluates the RBF exponential. After all support vectors are processed, an **accumulation and classification** step produces the output class label.

All arithmetic is performed in $\mathbb{Q}_{6.10}$ 16-bit signed fixed-point unless otherwise noted.

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## Fixed-Point Representation ($\mathbb{Q}_{6.10}$)

A 16-bit signed value $v$ is stored as the integer

$$\hat{v} = \lfloor v \cdot 2^{10} \rceil \in [-32768,\ 32767]$$

with the correspondence

$$v = \frac{\hat{v}}{1024}, \qquad \text{range: } [-32,\ +31.999], \qquad \text{LSB} = 2^{-10} \approx 0.001.$$

Multiplication of two $\mathbb{Q}_{6.10}$ values $\hat{a}$ and $\hat{b}$ produces a 32-bit intermediate result that is right-shifted by 10 bits to restore the $\mathbb{Q}_{6.10}$ format:

$$\widehat{a \cdot b} = \frac{\hat{a} \cdot \hat{b}}{2^{10}}.$$

The parameter $\gamma = 0.25$ is exactly representable: $\hat{\gamma} = 0.25 \times 1024 = 256 =$ `0x0100`.

## Distance Function (`dist`)

### Definition

For input feature vector $\mathbf{x} \in \mathbb{Q}_{6.10}^N$ and support vector $\mathbf{sv} \in \mathbb{Q}_{6.10}^N$ with $N = 256$ dimensions:

$$d^2(\mathbf{x}, \mathbf{sv}) \ = \ \sum_{i=0}^{N-1} (x_i - sv_i)^2 \, .$$

### Pipeline Implementation

The accumulator is a two-stage registered pipeline:

$$\delta_i = x_i - sv_i, \qquad s_i = \delta_i^2, \qquad D_k = \sum_{i=0}^{k} s_i.$$

Stage 1 (cycle $c$): compute and register $s_i = \delta_i^2$.
Stage 2 (cycle $c+1$): add $s_i$ to the running accumulator $D$.

After the $N$-th input pair is presented, **two drain cycles** are required before the result $D_{N-1}$ is valid:

- Drain cycle 1: $s_{N-1}$ propagates from the multiplier register to the adder input.
- Drain cycle 2: $s_{N-1}$ is added into $D$ and the result is registered.

Total cycles per SV distance: $N + 2 = 258$ at $\mathrm{LAT} = 1$; $\quad N \cdot \mathrm{LAT} + 2$ at general latency.

### Saturation

The accumulator is 32-bit, with a hardware maximum of $2^{32}-1$. The theoretical worst-case accumulated value for $N = 256$ and maximum feature magnitude $|x_i - sv_i| \leq 63.999$ is

$$D_{\max} = 256 \times (63.999)^2 \approx 1{,}048{,}544 \approx 2^{20},$$

which is well within the 32-bit range. The fixed-point overflow boundary is determined by the gamma multiply: after right-shifting by $2^{10}$, the maximum representable Q6.10 value from a 32-bit accumulator is

$$\frac{2^{32}}{2^{10}} = 2^{22}.$$

With $\hat{\gamma} = 2^8$, the accumulator would produce $\hat{p} > 2^{18}$ only if $D > 2^{20}$, which does not occur on real ECG data. The `ERR_DIST_OVERFLOW` flag is therefore a defensive sanity check rather than a condition expected in normal operation.

## Kernel Function (`horner`)

### Definition

The RBF (Gaussian) kernel is

$$K(\mathbf{x}, \mathbf{sv}) \;=\; e^{-\gamma\, d^2(\mathbf{x}, \mathbf{sv})}, \qquad \gamma = 0.25.$$

The argument $p = \gamma d^2$ ranges over $[0,\ \gamma D_{\max}] \approx [0,\ 262{,}136/1024] \approx [0,\ 256]$ in floating-point. In $\mathbb{Q}_{6.10}$ fixed-point, $\hat{\gamma} = 256 = 2^8$ and $\hat{d^2}$ reaches up to $\sim 2^{20}$ (saturating accumulator), so after the $\mathbb{Q}_{6.10}$ multiply (right-shift by $2^{10}$):

$$\hat{p} = \frac{\hat{\gamma} \cdot \hat{d^2}}{2^{10}} \leq \frac{2^8 \cdot 2^{20}}{2^{10}} = 2^{18}.$$

### Range Reduction

Direct Horner evaluation of $e^{-p}$ over $[0, 256]$ requires degree $\geq 8$ and overflows $\mathbb{Q}_{6.10}$ intermediate products. Instead, $p$ is split into integer and fractional parts:

$$I = \left\lfloor \frac{\hat{p}}{2^{10}} \right\rfloor \in [0,\ 255], \qquad F = \frac{\hat{p} \bmod 2^{10}}{2^{10}} \in [0,\ 1),$$

so that

$$e^{-p} = e^{-I} \cdot e^{-F}.$$

### LUT term: $e^{-I}$

A 256-entry read-only table stores

$$\texttt{lut}[I] = \left\lfloor e^{-I} \cdot 2^{10} \right\rfloor \in \mathbb{Q}_{6.10}, \qquad I = 0, 1, \ldots, 255.$$

For $I \geq 8$, $e^{-I} < 0.00034$, which rounds to zero in $\mathbb{Q}_{6.10}$, so the effective range is $I \in [0,\ 7]$. The LUT occupies $256 \times 16$ bits $= 4$ KB of on-chip flip-flop registers.

### Polynomial term: $e^{-F}$

For $F \in [0,\ 1)$, numerical analysis shows that degree 6 is the minimum Taylor polynomial that keeps the approximation error below the $\mathbb{Q}_{6.10}$ LSB of $2^{-10} \approx 0.001$ across the full range. The RTL defines up to 16 Taylor coefficients (`COEFF_00`–`COEFF_15`) and evaluates them via Horner's method. Defining signed coefficients $a_n = (-1)^n/n!$, the polynomial in standard form is:

$$e^{-F} \;\approx\; a_0 + a_1 F + a_2 F^2 + \cdots + a_d F^d,$$

where:

$$a_0 = 1, \quad a_1 = -1, \quad a_2 = 0.5, \quad a_3 = -0.1\overline{6}, \quad a_4 = 0.041\overline{6}, \quad a_5 = -0.008\overline{3}, \quad a_6 = 0.001\overline{388}.$$

Rather than evaluating each power of $F$ separately, Horner's method computes this as a recurrence starting from the innermost term:

$$h_d = a_d, \qquad h_k = a_k + F \cdot h_{k+1}, \qquad k = d-1,\ d-2,\ \ldots,\ 0.$$

The result is $h_0 = e^{-F}$. The alternating signs are carried in the $a_n$ coefficients, so every step of the recurrence is a simple multiply-add with no separate sign logic. Each step is one multiply and one add, requiring $d$ multiplications and $d$ additions total.

### Kernel Output

The two terms are combined by a $\mathbb{Q}_{6.10}$ multiply:

$$\hat{K} = \frac{\texttt{lut}[I] \cdot \hat{e}^{-F}}{2^{10}} \;\in\; [0,\ 1024],$$

where $\hat{K} = 1024$ corresponds to $K = 1.0$ (achieved when $d^2 = 0$, i.e. $\mathbf{x} = \mathbf{sv}$).

## Alpha Accumulation and OVR Decision

### Per-Class Accumulation

For each support vector $j = 0, \ldots, M-1$ with class label $\ell_j \in \{0, 1, 2, 3, 4\}$ and alpha coefficient $\alpha_j \in \mathbb{Q}_{6.10}$, the kernel output is weighted and accumulated into the corresponding class score:

$$f_c(\mathbf{x}) \;=\; \sum_{\substack{j=0 \\ \ell_j = c}}^{M-1} \alpha_j \cdot K(\mathbf{x}, \mathbf{sv}_j), \qquad c = 0, 1, 2, 3, 4.$$

The five accumulators $f_0, \ldots, f_4$ are 32-bit registers, updated after each SV kernel evaluation. Alpha coefficients are stored on-chip in a 500-entry register file (`alpha_table`[]), loaded via the Wishbone `ALPHA_WR` register before classification begins.

### Classification (Argmax)

After all $M = 500$ support vectors are processed, the predicted class is

$$\hat{y} = \arg \max_{c \in \{0,1,2,3,4\}} f_c(\mathbf{x}).$$

The argmax is implemented as a 5-way comparator tree and completes in one clock cycle. The result is registered on `class_out[2:0]` and exposed on GPIO.

## Operation Count Summary

| Stage | Operations per SV | SVs | Total |
|---|---:|---:|---:|
| `dist`: subtractions | 256 | 500 | 128,000 |
| `dist`: multiplications | 256 | 500 | 128,000 |
| `dist`: accumulations | 256 | 500 | 128,000 |
| `horner`: multiplications | 3 | 500 | 1,500 |
| `horner`: additions | 3 | 500 | 1,500 |
| Alpha multiply-accumulate | 1 | 500 | 500 |
| **Total per beat** | | | **387,500** |

At $N = 256$ and $M = 500$, the design is strongly **memory-bound**: arithmetic intensity $AI \approx 1.5$ OPs/byte (LUT preloaded), placing the operating point well below the compute ceiling of the 40 MHz fixed-point datapath.

## FSM Cycle Budget (LAT = 3)

$$\begin{aligned} T_{\text{beat}} &= T_{\text{load}} + M \cdot (T_{\text{dist}} + T_{\text{kernel}}) + T_{\text{classify}} \\ &= N \cdot \text{LAT} + M\,(N \cdot \text{LAT} + 2 + 18) + 1 \end{aligned}$$

$$= 256 \times 3 + 500 \times (256 \times 3 + 20) + 1 = 768 + 500 \times 788 + 1 = 394{,}769 \text{ cycles/beat.}$$

At 40 MHz: $394{,}769/40 \times 10^6 \approx 9.87$ ms/beat, consistent with the measured 9.7 ms (the small difference reflects pipeline overlap between LOAD and COMPUTE).